

Groups

Chapter 1. Groups

This material follows the standard treatment in a [1].

Exercise 1.1. Consider an exercise.

See [Exercise 1.0](#).

Exercise 1.2. funn

Solve [Exercise 1.1](#).

§1.1 Fields

We begin by establishing the properties of fields mapping vector components.

Definition 1.3. A function $T : V \rightarrow W$ between two vector spaces over the same field F is called a **linear transformation** if it satisfies:

1. Additivity: $T(u + v) = T(u) + T(v)$ for all $u, v \in V$
2. Homogeneity: $T(cv) = cT(v)$ for all $c \in F$ and $v \in V$

[Definition 1.2](#).

Definition 1.4. ...

§1.2 Degree of an extension

Theorem 1.5. Let V and W be vector spaces, where V is finite-dimensional. If $T : V \rightarrow W$ is a linear map, then:

$$\dim(\text{null } T) + \dim(\text{range } T) = \dim V$$

§1.3 Exercises

Exercise 1.6. Let $T : \mathbb{R}^3 \rightarrow \mathbb{R}^2$ be defined by $T(x, y, z) = (x + y, z)$. Find a structured matrix representation M_T and verify the output dimension explicitly:

$$M_T = \begin{pmatrix} 1 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

Exercise 1.7 (Identity Dimension Mapping).

Let $T : \mathbb{R}^3 \rightarrow \mathbb{R}^2$ be defined by $T(x, y, z) = (x + y, z)$. Find the dimension of the null space.

Solution:

By inspecting the matrix row parameters:

$$M_T = \begin{pmatrix} 1 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

The rank is clearly 2 because the two rows are linearly independent. By Rank-Nullity, $\dim(\text{null } T) = 3 - 2 = 1$.

Note 1.8. Remember that every basis is linearly independent.

Warning 1.9. Do **not** confuse image and codomain.

Example 1.10. Let

$$T(x, y) = (x + y, y).$$

Proof. We proceed by induction...

□

Remark 1.11. The converse is false.

History 1.12. The Rank-Nullity theorem appeared in the nineteenth century.

Lemma 1.13. lemma statement

Proposition 1.14. proposition statement

Claim 1.15. claim statement

Corollary 1.16. corollary statement

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Bibliography

[1] D. S. Dummit and R. M. Foote, *Abstract Algebra*, 3rd ed. Wiley, 2004.